

43. Resource Loading: onload aur onerror

Scripts, images, iframes jaise external resources ke loading ko track karne ke 2 events:

- `onload`: successful load
- `onerror`: error hua

Script load karna

```
js
let script = document.createElement('script');
script.src = "my.js";
document.head.append(script);
```

Us script ke functions chalane ke liye load hone ka wait karna padta hai.

`script.onload`

Script load **aur execute** hone ke baad chalta hai.

```
js
script.onload = function() {
  alert(_VERSION); // script ka variable use kar sakte ho
};
```

`script.onerror`

Load fail hone par (jaise 404, server down):

```
js
script.onerror = function() { alert("Error loading " + this.src); };
```

Dhyan: **HTTP error details nahi milte** (404 hai ya 500), sirf itna ki load fail hua.

Important: `onload/onerror` sirf **loading** track karte hain. Script chalte waqt jo errors aaye unke liye nahi. Script load ho gayi to `onload` chalega chahe usme bugs hon. Runtime errors ke liye `window.onerror` use karo.

Baaki resources

`load` aur `error` har us resource ke liye chalte hain jiska external `src` ho.

```
js
```

```
let img = document.createElement('img');
img.src = "train.gif"; // img document me daalne se nahi, src milte hi load s
img.onload = ...;
img.onerror = ...;
```

- Zyaadatar resources document me add hone par load shuru karte hain, `<img>` **exception** hai (src milte hi).
- `<iframe>`: `onload` hamesha chalta hai (success ho ya error), historical wajah se.

Crossorigin policy

Ek site ki script dusri site ka content nahi padh sakti (origin = protocol + domain + port). Ye rule resources par bhi lagta hai.

Dusre domain ki script me error ho to `window.onerror` ko sirf "**Script error.**" milta hai, details (line, stack) chhupi rehti hain. Details chahiye to:

<code>crossorigin</code> attribute	Matlab
Koi nahi	Access mana
<code>crossorigin="anonymous"</code>	Server <code>Access-Control-Allow-Origin</code> (*) ya hamara origin) bheje to allowed, cookies nahi jaati
<code>crossorigin="use-credentials"</code>	Server ko <code>Access-Control-Allow-Origin</code> (hamara origin) + <code>Access-Control-Allow-Credentials: true</code> dena hoga, cookies bhi jaati hain

html

```
<script crossorigin="anonymous" src="https://other-site.com/error.js"></script>
```

Summary

- `load`: success, `error`: fail.
- `<iframe>` ka `load` hamesha trigger hota hai.
- `readystatechange` bhi chalta hai par kam use hota hai.

Task: `preloadImages(sources, callback)`

Saari images ko load karke, sab ke load **ya error** hone par `callback()` chalao. Algorithm: har source ke liye `img` banao, `onload/onerror` lagao, dono me counter badhao, counter == sources ki length to `callback()`.